

WEEK 5, BEFORE THE FIRST PASS

Week 5 pre-read.

About 30 minutes. Graded complete or incomplete. Do this before the first pass.

This is a quick look ahead, not a full read. Look at the headings and the figures, answer the questions below in a few words, and move on. You will learn all of it properly in the walkthroughs, and it will make much more sense because you have already seen it once. **This week covers three chapters: Chapter 14 (reflexes), Chapter 10 (the spinal cord, cerebrospinal fluid and the blood–brain barrier), and Chapter 11 (sensory physiology).** Every section below has a part for each.

Where to look

Read closely, Silverthorn Chapter 14, Integrative

Physiology I: Control of Body Movement: only the part on neural reflexes and skeletal muscle reflexes. That covers how reflexes are sorted, the muscle spindle, the Golgi tendon organ, and the withdrawal and crossed extensor reflexes. The rest of Chapter 14 is next week.

Read closely, Silverthorn Chapter 10, The Central Nervous

System: the parts on the membranes, cerebrospinal fluid and the blood–brain barrier, and the whole section on the spinal cord. In the section on the brain, read the part on the brain stem and look over Table 10.1, the cranial nerves, which you use in lab. In the section on brain function, read only the parts on how the spinal cord and brain handle sensory information and on motor output. The rest of the chapter, on sleep, memory, language and emotion, is not part of this week.

Then, Silverthorn Chapter 11, Sensory Physiology. Read the first part, on the general properties of sensory systems, and the part on the somatic senses, including pain. For the special senses, vision, hearing, balance, taste and smell, only skim: look at the figures of the eye and the ear and read the chapter's Visual Summary. The walkthroughs teach what you need from them.

Chapter numbers are for the 9th edition. Other editions number things differently, so go by the chapter title and the topic, and use the search in your eText to find it.

Figures to look at closely

These are the most important figures for this week. Open each one in your eText, read its labels, and look for the thing named after it.

Chapter 14, reflexes

- **Table 14.1, Classification of Neural Reflexes.** The four ways a reflex can be sorted.
- **Fig. 14.1, Neural Reflexes.** Count the synapses in each of the three reflexes.
- **Fig. 14.2, Muscle Spindles and Alpha-gamma Coactivation.** Find which part of the spindle has no myofibrils, and where the gamma motor neuron attaches.
- **Fig. 14.3, Golgi tendon organs.** Find what squeezes the nerve endings when the muscle contracts.
- **Fig. 14.5, The patellar tendon (knee jerk) reflex.** Follow both efferent paths: one to the quadriceps and one to the hamstrings.
- **Fig. 14.6, The crossed extensor reflex.** Find which muscles contract and which are inhibited in each leg.

Chapter 10, the spinal cord and protecting the brain

- **Fig. 10.6, Organization of the spinal cord.** Find the posterior horn, the anterior horn, and where the ascending and descending tracts run.
- **Fig. 10.4, Cerebrospinal Fluid.** Trace the fluid from where it is made to where it goes back into the blood.
- **Fig. 10.5, The blood-brain barrier.** Find the two kinds of cells that help keep the capillary tight.
- **Table 10.1, The Cranial Nerves.** You will use this in lab.

Chapter 11, sensory physiology

- **Fig. 11.2, Receptive fields.** See why two points close together are felt as one on the arm but as two on a fingertip.
- **Fig. 11.7, Receptor adaptation.** Compare the action potentials from a tonic and a phasic receptor.
- **Fig. 11.8, Somatosensory pathways.** Find where pain crosses the midline and where fine touch crosses.
- **Fig. 11.10, Sensory receptors in the skin.** Find the receptor that senses steady pressure and texture.
- **Fig. 11.11, Referred pain.** Find where pain from the heart is felt.
- **For the special senses, just look at:** Fig. 11.24 (the eye), Fig. 11.26 (optics of the eye), Fig. 11.18 (signal transduction in hair cells) and Fig. 11.21 (equilibrium).

What this reading should help you do

Read each goal and pick the answer that fits you right now. There is no wrong answer here.

Reflexes and the spinal cord (Chapters 14 and 10)

Name the five parts of a reflex arc, and sort a reflex as somatic or autonomic, spinal or cranial, innate or learned, and monosynaptic or polysynaptic.

☐ This is new to me ☐ I have heard of it

☐ I could explain it now

Explain what the muscle spindle and the Golgi tendon organ each detect, and what their reflexes do.

☐ This is new to me ☐ I have heard of it

☐ I could explain it now

Trace the withdrawal and crossed extensor reflexes, and explain why they need interneurons.

☐ This is new to me ☐ I have heard of it

☐ I could explain it now

Tell an upper motor neuron problem from a lower motor neuron problem by the reflexes and muscle tone.

☐ This is new to me ☐ I have heard of it

☐ I could explain it now

Describe what the gray matter and white matter of the cord hold, which pathways carry touch, pain and movement, and where cerebrospinal fluid and the blood-brain barrier fit in.

☐ This is new to me ☐ I have heard of it

☐ I could explain it now

Sensory physiology (Chapter 11)

Explain how a receptor turns a stimulus into a signal, and how the brain knows what kind of stimulus it was, where, and how strong.

☐ This is new to me ☐ I have heard of it

☐ I could explain it now

Explain how pain is carried and how it can be turned down.

☐ This is new to me ☐ I have heard of it

☐ I could explain it now

Name what each special sense detects, and the main structures of the eye and the ear.

☐ This is new to me ☐ I have heard of it

☐ I could explain it now

Find it

Each answer is in the part of the chapter named. A few words is enough.

Reflexes and the spinal cord (Chapters 14 and 10)

Name two ways reflexes can be sorted, and give one example reflex for each kind.

In Chapter 14, the part on neural reflexes. Look at Table 14.1.

A muscle spindle and a Golgi tendon organ: where is each one, and what does each one sense?

In Chapter 14, the part on skeletal muscle reflexes. Search your eText for muscle spindle. Look at Figs. 14.2 and 14.3.

What do gamma motor neurons do for the muscle spindle?

Same part of Chapter 14. Search for gamma motor neuron. Look at Fig. 14.2, part (c).

When you step on something sharp, what does the other leg do, and what is that reflex called?

Same part of Chapter 14. Search for crossed extensor. Look at Fig. 14.6.

What is the difference between a horn, a root, a tract and a column of the spinal cord? Which root carries sensory information in, and which carries motor commands out?

In Chapter 10, the section on the spinal cord. Look at Fig. 10.6.

Name the three membranes around the brain from the skull inward. Where does cerebrospinal fluid flow, where is it made, and where does it go back into the blood?

In Chapter 10, Anatomy of the Central Nervous System, the parts on the membranes and on cerebrospinal fluid. Look at Figs. 10.3 and 10.4.

What makes brain capillaries tight, which two kinds of cells help keep them that way, and why can L-dopa reach the brain when dopamine cannot?

In Chapter 10, the part on the blood-brain barrier. Look at Fig. 10.5.

Look at Table 10.1, the cranial nerves. Which nerves are sensory only, and which nerve controls pupil size and the shape of the lens?

In Chapter 10, the section on the brain, the part on the brain stem. Look at Table 10.1.

Sensory physiology (Chapter 11)

What is an adequate stimulus? Name the four groups of sensory receptor in Table 11.2 and give one example of each.

In Chapter 11, the section on general properties of sensory systems. Look at Table 11.2.

What is a receptive field, and why can you tell two points apart better on a fingertip than on your back?

In Chapter 11, the part on receptive fields and two-point discrimination. Look at Fig. 11.2.

Why can pain from the heart be felt in the left arm?

In Chapter 11, the part on pain. Search for referred pain. Look at Fig. 11.11.

Words to know

Find each term in the reading. Write what it means in your own words, not the book's. If you are working on paper, add a quick sketch. If you are typing, give one example instead.

Reflex arc

Interneuron

Receptor potential

Labeled line

Receptive field

Adaptation

Nociceptor

Muscle tone

Upper motor neuron

Blood-brain barrier

Before you start the walkthroughs

Which of this week's topics looks hardest to you right now, and why?

Write one question you want the walkthroughs to answer.

Turn it in

Save this page as a PDF with the button below and upload it to the Week 5 pre-read assignment in Canvas. If you worked on paper, upload a photo of your pages instead.